PROJECT PROPOSAL: SELF DEFEND

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ECE 424, Fall 2026

Due: Monday, October 5, 2026, 11:59 PM New Jersey time

Selected topic OR paper: [Self Defend: LLMs Can Defend Themselves against Jail-breaking in a Practical Manner

Replace the bracketed prompts with your answers and submit a one-page PDF. Concise sentences or bullets are welcome. Completed experimental results are not required.

1. RESEARCH QUESTION AND SAFE SCOPE 5 points

Question / claim: Can a shadow LLM jailbreak detector reduce the success of jailbreak attacks against a targeted LLM without significantly increasing latency or blocking harmless prompts?

Setting and threat model: We will evaluate a text-based target LLM protected by a separate shadow defense LLM. The asset in this case is the safety allignment of the model and the attacker has a black box prompt access and will intend to induce a prohibited output with jailbreak prompts. The attacker, however, cannot modify weights, server side code, or the defense.

Resources and authorization: Any testing will use only locally hosted/open models or approved API (all in an authorized environment). No attacks that are performed will be against any third party system. Compute/API use will be capped depending on the constraint of the project

2. BASELINE, CONTRIBUTION, AND EVALUATION 5 points

Prior work and baseline: We will attempt to reproduce a simplified SELFDEFEND setup for mentioned in the paper. A target LLM generates the answer while a show LLM checks the same input. We will then compare an unprotected target baseline with the paper's direct content detector (PDirect) and intent based detector (PIntent), reusing any public benchmark prompts and available model implementations.

Bounded contribution: We will be testing whether a lower cost shadow model can perserve most of SELFDEFENDs security benefit. The study will compare detector size as well as detection strategy to characterize the tradeoff of jailbreak resistance, latency, and false positives

Experiment and metric: We will evaluate apporximately 100 jailbreak prompts and 100 benign prompts across 4 configurations (Target-only, Pdirect, Pintent, and a lightweight shadow model). The primary metric would be based on the jailbreak attack success rate (ASR). Secondary metrics would include benign prompt pass rates and latency additions. Configuration choices will be based on a developmental subset and final results will use a held-out evaluation set

3. INDIVIDUAL OWNERSHIP AND WORK ALLOCATION 5 points

Member name	**Owned technical component**	**Owned experiments (at least two)**	**Responsible report sections / subsections**
Samarth Kaushik	[Target LLM + Baseline Harness]	[1: Baseline ASR; 2: Benign Prompt Test]	[Threat Model; Baseline]
Sharon Melendez	[Pdirect Shadow Defense]	[1: Pdirect Jailbreak Test; 2: Latency Test]	[Architecture + Pdirect Results]
Yasmin Shahlamedina	[Pintent & Lightweight Defense]	[1: Pintent Comparison; 2: Model Size Stress Test]	[Contribution; Comparative Analysis]

Each member needs a component, at least two experiments, and report responsibilities. Update this table if responsibilities change. Keep workloads balanced; this table informs end-of-semester individual grading. For an approved pair or the 539 individual project, remove unused member rows and author names.